

Inositol Hexakisphosphate Inhibits Osteoclastogenesis on RAW 264.7 Cells and Human Primary Osteoclasts

Background Inositol hexakisphosphate (IP6) has been found to have an important role in biomineralization and a direct effect inhibiting mineralization of osteoblasts in vitro without impairing extracellular matrix production and expression of alkaline phosphatase. IP6 has been proposed to exhibit similar effects to those of bisphosphonates on bone resorption, however, its direct effect on osteoclasts (OCL) is presently unknown. **Methodology/Principal Findings** The aim of the present study was to investigate the effect of IP6 on the RAW 264.7 monocyte/macrophage mouse cell line and on human primary osteoclasts. On one hand, we show that IP6 decreases the osteoclastogenesis in RAW 264.7 cells induced by RANKL, without affecting cell proliferation or cell viability. The number of TRAP positive cells and mRNA levels of osteoclast markers such as TRAP, calcitonin receptor, cathepsin K and MMP-9 was decreased by IP6 on RANKL-treated cells. On the contrary, when giving IP6 to mature osteoclasts after RANKL treatment, a significant increase of bone resorption activity and TRAP mRNA levels was found. On the other hand, we show that 1 μ M of IP6 inhibits osteoclastogenesis of human peripheral blood mononuclear cells (PBMNC) and their resorption activity both, when given to undifferentiated and to mature osteoclasts. **Conclusions/Significance** Our results demonstrate that IP6 inhibits osteoclastogenesis on human PBMNC and on the RAW264.7 cell line. Thus, IP6 may represent a novel type of selective inhibitor of osteoclasts and prove useful for the treatment of osteoporosis.